Refactor CS/SE majors: a strawperson proposal (GKT/KL)

- Maintain a lean foundation sequence (163, 170, 264, 271) - i.e. the kernel, inspired by Mach
- Keep 272 (Data Structures II) but add 280 (Introduction to AI/Fundamentals) - gateway course
- Consider merging CS and SE. CS already emphasizes both.
- Remove restricted electives – schedule key 300-level courses by area:
  - Theory, Languages and systems, AI, SE, Applications
  - Life success (aka soft skills, sys admin, etc.)
- Eliminate as many 300/400 combinations *and common course* names as possible/sensible
  - 300 "X" for primarily undergrad
  - 400 "Adv X" for primarily grad – with option for adv. undergrads to take it
  - There are counterexamples, e.g., 353/453, but we should think how to streamline these, too.
- Viz on next slide (using data from George's advising app, https://advising.cs.luc.edu.
  - This viz is just a proof of concept and can be further refined in Curricular Retreat.
  - Unlikely to push any major initiative this year but hopefully the next administration (chair wise) will consider it
- Cybersecurity and IT are both out of scope in this strawperson proposal. They have distinct audiences and expectations.

A kernel, then layers

Proposed CS BS major: 60 credits (61 today). Core, language and CAS requirements are unchanged.

DRAFT FOR DISCUSSION

Final courses and areas are subject to review.

Experience

15 credits

depth + practicum / capstone

3 more 300-level courses

any area · 9 cr

Practicum / Capstone

6 cr total · COMP 398 Independent Study + COMP 391 Internship

every 300-level course offers a capstone experience

Breadth

21 credits

ethics + 1 from each of 6 areas

Theory

pick one · 3 cr

- **363** Algorithms
- **376** Automata
- **350** Applied Crypto
- **331** Crypto Math
- **328** Coding Theory
- **309** Numerical Meth.

Languages & Systems

pick one · 3 cr

- **310** Operating Systems
- **362** Architecture
- **371** Prog. Languages
- **335** Formal Methods
- **353** Database Prog.
- **343** Networks

+ 16 more

Software Engineering

pick one · 3 cr

- **330** Software Eng.
- **313** OO Design
- **332** Requirements
- **370** Software Quality
- **312** Open Source

+ 7 more

Applications

pick one · 3 cr

- **323** Game Design
- **341** HCI
- **380** Graphics
- **306** Data Mining
- **358** Big Data
- **387** Deep Learning

+ 10 more

AI

pick one · 3 cr

- **378** Artificial Intel.
- **379** Machine Learning

these two only

Life Skills

pick one · 3 cr

- **377** IT Proj. Mgmt
- **395** Career Growth+
- **397** Research Meth.+
- **390** Broadening Part.+
- **314** Problem Solving+

+ 1 more

Ethics

required · 3 cr

- **317** Ethics

everyone takes it

Kernel

24 credits

required of everyone

MATH	COMP 163	COMP 170	COMP 264	COMP 271	STAT 203	COMP 272 or 280*
131/132 or 161/162	Discrete Structures	Object-Oriented Programming	Computer Systems	Data Structures I	Probability & Statistics	Theory track or AI track
Calculus	Foundation sequence				Statistics	Advanced 200-level

* COMP 280 is a proposed new course (assumed 3 credits) · + 1-credit course · Examples drawn from the 2026–27 catalog; each tile lists a sample of its area.